Fikile Mkhize 0210210785080

0781087624

13 August 2026

DEMONSTRATE AN UNDERSTANDING OF TESTING IT SYSTEMS AGAINST GIVEN SPECIFICATIONS

US 14908

SECTION A: FORMATIVE ASSESSMENT

QUESTION 1

When selecting a test procedure we must consider:

1. Purpose of the test - Is it for functionality, security, performance, or user acceptance?

2. Type of system - Hardware vs software vs network requires different tools

3. Risk and impact - High-risk systems like AVBOB claims system need more rigorous testing

4. Compliance requirements - FAIS, POPIA, and company policies must be met

5. Resources available- Time, budget, staff, and testing tools

6. Test environment - Should mimic live environment but not affect production data

7. User requirements - Test must cover what end-users actually need

8. Test data - Must use realistic but non-sensitive data

9. Documentation - Procedure must be clearly documented and repeatable

10. Cost vs benefit - Ensure the testing adds value and doesn't waste resources

QUESTION 2

1. Define objectives - What are we testing? E.g. Test if new AVBOB policy system can process claims

2. Identify scope - Which modules, users, and data are included

3. Develop test plan - Include timelines, roles, and responsibilities

4. Gather requirements - Get business rules and user specs

5. Prepare test environment - Set up test servers, not live data servers create test cases - Step-by-step actions and expected results. E.g. "Login with valid ID → Access granted"

7. Prepare test data - Use dummy client info that follows POPIA

8. Assign roles - Who is tester, who approves, who fixes bugs

Fikile Mkhize

9. Get sign-off- Manager approves plan before testing starts

10. *Backup systems* - Ensure data can be restored if test fails

QUESTION 3

Using AVBOB as example:

Hardware Testing:

1. Installation check- PCs, printers, scanners are checked on delivery

2. Functionality test - Test if hardware works: e.g. test printer prints claim forms

3. Performance test - Check speed and reliability under load

4. Compatibility test- Ensure new hardware works with AVBOB CRM

Software Testing:

1. Unit testing - Developers test individual parts of the system

2. system testing- Test full AVBOB policy system end-to-end

3. User Acceptance Testing - UAT- Contact centre agents test if system meets daily needs

4. Regression testing - After updates, test that nothing else broke

5. Security testing - Check access controls and data

QUESTION 4

1. Clear and documented - Steps are easy to follow and repeat

2. Comprehensive - Covers all requirements and possible scenarios

3. Traceable - Each test links back to a business requirement

4. Measurable - Has pass/fail criteria and records results

QUESTION 5

1. Accuracy- Data must reflect real results, no guessing

2. Completeness- Record all test steps, inputs, and outputs

Fikile Mkhize

3. Confidentiality- Use dummy data, follow POPIA. No real client info

4. Consistency - Use same format for all tests

5. Traceability- Link data to specific test case and date

6. Security - Store data in protected folders with access control

7. Timeliness - Record immediately after test, not later

8. Relevance - Only record data needed for the test objective

9. Backup - Save in 2 places to avoid loss

10. Review - Supervisor should verify data before sign-off

QUESTION 6

1.Purpose and scope- What system is being tested and why

2. Objectives- What we want to achieve

3. Test items - Hardware, software, modules to be tested

4. Approach/methodology- Manual vs automated testing

5. Resources- People, tools, environment needed

6. Schedule - Dates and timelines

7. Roles and responsibilities- Who does what

8. Test cases - What will be tested

9. Risk and assumptions - What could go wrong

10. Sign-off criteria - When is testing considered complete

QUESTION 7

1. Understand requirements- Read business and technical specs

2. Develop test strategy - Decide type of testing needed

3. Create test plan- Document scope, schedule, resources

4. Set up test environment - Separate from production

5. Prepare test data - Create dummy but realistic data

6. Write test cases- Detailed steps and expected outcomes

Fikile Mkhize

7. Review test cases - Peer review for gaps

8. Identify tools- Testing software, bug tracking tools

9. Train testers - Ensure team understands procedure

10. Get approval - Manager signs off plan

11. Backup- Backup current system before testing

12. Communication - Inform all stakeholders of test dates